Project 2: Singapore Rainfall Analysis and Prediction

The Historical Rainfall across Singapore datasets, available at **https://data.gov.sg/collections/2279/view (https://data.gov.sg/collections/2279/view)** , provide rainfall measurements recorded at five-minute intervals by monitoring stations across Singapore. Use the annual datasets from **2017 to 2024**. Each record includes the observation time, station information, and rainfall amount (mm).

The project involves structured data processing and machine learning using Apache Spark. You may use any programming language supported by Spark. The two tasks are as follows:

- **Task 1: Explore weather patterns.** Explore the weather patterns reflected in the rainfall data and how they change within a year and across the years 2017–2024. Present and discuss your findings using appropriate analysis and visualizations.
- **Task 2: Predict rainfall.** Build a machine learning model that estimates the probability that a monitoring station will accumulate more than a given amount of rainfall in the next 30 minutes. The model should take the station, prediction time, and rainfall threshold as inputs. Select suitable features and evaluate your model using Spark MLlib.

The project requires you to write a report describing your approach and results. In the report, you must

- Explain how you cleaned the data, investigated weather patterns and their changes, selected features, and developed your machine learning model.
- Describe your computing environment (e.g., a local machine, Google Cloud, or Databricks), including the relevant VM or cluster configuration.
- Present and discuss the results of both tasks, preferably with figures and tables, and evaluate your model's predictive performance.
- Declare any AI usages.
- Discuss any additional work, observations, limitations, or insights you would like to share.

This is again a group project, and each group can have at most three people. Please submit the following materials to Canvas no later than **18 Oct 2026**:

- Your complete source code and a README explaining how to run it.
- Your trained model.
- A PDF report listing the names and student IDs of all group members.